Learning Dynamics

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Stormglass

Big questions

- How a model generalize
- Internal structure formation during training
- Emergent capabilities
- Control levers of the training process
- How loss, architecture, data, optimizer determinate how internal structure is learned throughout training

How: making mathematical models of learning

- Tractable toy models: deep linear networks, shallow non linear networks, polynomial activations
- Take limits: depth, width, data, training time
- Exploit symmetries: quotient them out, find conserved quantities
- Identify useful coarse grained variables: singular values of weights, learning coefficient, correlation functions
- Work in specific regimes: lazy vs rich
- Validate with tight feedback loops: run experiments, symbolic derivations, mathematical proofs

Safety motivations

- A central question in AI safety is wether the training process select a harmful model (scheming, sycophantic, spiteful...)
- Predict if a model is going to be harmful in high stakes and out of distribution environment while behavioural or average case interpretability evals cannot be trusted
- Develop stronger interpretability methods by decomposing throughout training
- We want a mathematical model that can beat average methods at predicting harmful behaviour
- More ambitiously we would like to find control levers of the training process to be confident that training will select a safe model

Setup

Deep neural network

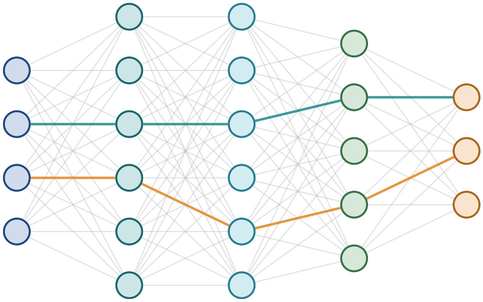
Let parameter $\theta \in \Omega$, input $x \in \mathbb{R}^{d_0}$, weight matrices $W_l \in \mathbb{R}^{d_l \times d_{l+1}}$, activation function σ:

$$h_l(\theta, x) = \sigma(W_l h_{l-1} + b_l)$$

A neural network is a function f such that:

$$f(\theta, x) = W_L h_{L-1}(\theta, x)$$

Other architectural components: residual stream, tokenization, embeddings, attention for sequence modeling, layer normalization...



Deep Linear Network

Function is linear in data, non linear in parameters:

$$f : \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$

$$(x, W_1, ..., W_L) \mapsto W_L ... W_1 x = Wx$$

d_0 is the input dimension; d_ℓ, $1 \leq \ell \leq L$, is the width of layer ℓ. Define the product map:

$$\mu(\theta) := W_L ... W_1 = W$$

Where W is the student matrix.

Stochastic gradient descent

Learn a function g from observing $y = g(x)$. Initialize neural weights at $\theta_0 \sim N(0, \sigma^{-\gamma})$ with initialization scale γ.

SGD update

Consider dataset of size N. At each time step sample batch b of size B. Empirical loss and batch loss relation:

$$\mathcal{L}(\theta) = \langle \mathcal{L}(\theta; b) \rangle_b$$

SGD updates is:

$$\theta_{k+1} = -\eta \nabla_{\theta_k} \mathcal{L}(\theta_k) + \eta \xi(\theta_k, b_k)$$

With gradient noise

$$\xi(\theta, b) := \nabla \mathcal{L}(\theta_k) - \nabla \mathcal{L}(\theta_k; b_k)$$

Loss function $\mathcal{L}$ can for example be mean square error or cross entropy loss

Data, loss function, gradient flow

Quadratic population loss and gradient flow

Let M be the teacher matrix, with $y = Mx$ and $x \sim \mathcal{N}(0, I)$. The input-output covariance is $\Sigma_{YX} = M$.

$$\mathcal{L}(\theta) := \frac{1}{2}\|M - W_L \cdots W_1\|_F^2 = \frac{1}{2}\|M - W\|_F^2$$

Initialize $\theta_0 \sim \mathcal{N}(0, \sigma^{-\gamma})$, where γ controls the initialization scale. Then gradient flow is

$$\dot{W}_\ell = -\nabla_{W_\ell}\mathcal{L}(\theta).$$

Frobenius norm: $\|A\|_F := \sqrt{\mathrm{Tr}(A^\top A)}$.
Global minimizers achieve zero loss: $\exists \theta^\star \in \Omega,\ \mu(\theta^\star) = W^\star = M$.
The structure of the data is encoded by Σ_{YX}.

Geometry

Global minima and saddle point

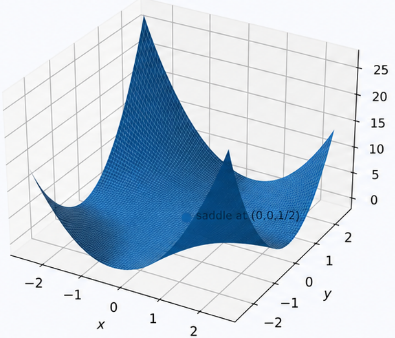
Loss: $\ell_1(x, y) = \frac{1}{2}(xy - 1)^2$

Global minima: $\{(x, y) : xy = 1\},\ \ell_1 = 0$

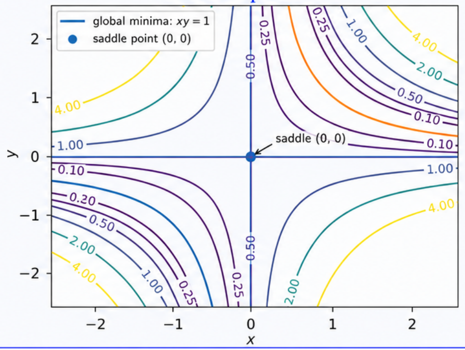
Saddle point: $(0, 0),\ \ell_1(0, 0) = \frac{1}{2}$

Strict saddle: along $x = y = t$ the loss decreases near 0, while along $x = -y = t$ the loss increases near 0

3D loss surface



Contour plot



Global minima form the hyperbola $xy = 1$, while $(0, 0)$ is a **strict saddle point**.

Deep learning without poor local minima

Assume that Σ_X and Σ_{YX} are full rank, and that Σ_{YX} has d_L distinct singular values.

No non-global local minima

The loss function of a DLN of depth L is non-convex and non-concave, and every critical point that is not a global minimum is a saddle point.

The training dynamics cannot get stuck in spurious local minima.
Reference: *Kawaguchi, Kenji. "Deep learning without poor local minima." Advances in neural information processing systems 29 (2016)*

Parametrization of critical points

Teacher SVD: $M = U \operatorname{diag}(s_1, \ldots, s_{r_m}, 0_{d_L - r_m}) V^\top$.

Saddle points

Let $\theta \in \Omega$ be a critical point of rank r. Then there exists a subset S of left singular vectors of M and a projector $P_S = U_S U_S^\top$ onto their span such that:

$$\mu(\theta) = P_S M = U_S \operatorname{diag}(s_1, \ldots, s_r, 0_{d_L - r}) V^\top$$

The saddle points are given by a subset of singular values and (left) singular vectors of the teacher M.

Reference: *Achour, El Mehdi, François Malgouyres, and Sébastien Gerchinovitz. "The loss landscape of deep linear neural networks: a second-order analysis." Journal of Machine Learning Research 25.242 (2024): 1-76.*

Symmetries

Let $G_{\mathbf{d}} := \prod_{\ell=0}^{L} GL_{d_\ell}$ act on Ω via:

$$(W_1, \ldots, W_L) \mapsto (g_1 W_1 g_0^{-1},\ g_2 W_2 g_1^{-1},\ \ldots,\ g_L W_L g_{L-1}^{-1}),$$

Equivariance of the multiplication map

Since $W_L g_{L-1}^{-1} g_{L-1} W_{L-1} \cdots g_1^{-1} g_1 W_1 = W_L \cdots W_1$, the multiplication map μ is $G_{\mathbf{d}}$-equivariant and G_h-invariant:

$$\mu(g \cdot \theta) = g_L W g_0^{-1}, \qquad \mu(g_h \cdot \theta) = \mu(\theta).$$

A DLN is degenerate: infinitely many DLNs can map to the same student matrix.

It is natural to study solutions up to G_d symmetries. By equivariance of μ, G_d acts on the determinantal variety

$$\Sigma_d^r := \{\theta \in \Omega \mid \mathrm{rank}(\mu(\theta)) = r\}.$$

Stratification of the fiber of the parameter function map

Orbits are parameterized by rank patterns

The G_d-orbits in Σ_d^r are indexed by rank patterns $\rho_{ij} = \mathrm{rank}(W_j \ldots W_i)$:

$$\Sigma_d^r = \bigsqcup_{\rho_{ij}} \mathcal{O}_\rho$$

Two DLN are in the same G_d-orbit if and only if they have the same rank pattern
Since $\Sigma_d^r = G_{\mathrm{out}} \cdot \mu^{-1}(W)$, this gives a combinatorial description of the fiber of the product map $\mu^{-1}(W)$

Rank patterns are natural observables from geometric invariants for DLNs.

Link with SLT

Codimension of the global minima

For a teacher matrix M of rank r_m, we have:

$$\begin{aligned}\operatorname{codim}(\mu^{-1}(M)) &= \operatorname{codim}(\Sigma_d^{r_m}) + r_m(d_0 - d_L - r_m)\\ &= 2\,\mathrm{RLCT}(\mathcal{L}^{-1})(0).\end{aligned}$$

In general, we only have:

$$\mathrm{RLCT}(\mathcal{L}^{-1})(0) \leq \frac{1}{2}\operatorname{codim}_{\mathrm{Rep}_d}(\mathcal{L}^{-1}(0)).$$

Ref: *Lehalleur, Simon Pepin, and Richárd Rimányi. "Geometry of fibers of the multiplication map of deep linear neural networks." arXiv preprint arXiv:2411.19920 (2024).*

Discussion

Takeaways

Under full-rank and nondegenerate-data assumptions:

- Critical points of DLNs are either saddles or global minima
- Gauge symmetries mean there is a whole algebraic variety of degenerate minima
- Geometric invariants give observables and a combinatorial description of the parameter function map and the global minima

References

- Kenji Kawaguchi. Deep learning without poor local minima *Advances in neural information processing systems*, 29, 2016.
- El Mehdi Achour, François Malgouyres, and Sébastien Gerchinovitz. The loss landscape of deep linear neural networks: a second-order analysis. *Journal of Machine Learning Research*, 25(242):1–76, 2024.
- Anna Choromanska, Mikael Henaff, Michael Mathieu, Gérard Ben Arous, and Yann LeCun. The loss surfaces of multilayer networks. In *Artificial intelligence and statistics*, pages 192–204. PMLR, 2015.
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- Chen, Po and Jiang, Rujun and Wang, Peng. A Complete Loss Landscape Analysis of Regularized Deep Matrix Factorization, arXiv preprint arXiv:2506.20344, 2025.

Dynamics

Gradient flow and NTK dynamics

Small-learning-rate limit of gradient descent:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \qquad \rightsquigarrow \qquad \dot{\theta}(t) = -\nabla_\theta \mathcal{L}(\theta(t))$$

Neural Tangent Kernel (NTK):

$$K(x, x', \theta) = \nabla_\theta f(\theta, x) \nabla_\theta f^\top(\theta, x) = \sum_{i=1}^{p} \partial_{\theta_i} f(\theta, x) \partial_{\theta_i} f^\top(\theta, x)$$

For a quadratic loss on a dataset (x_i, y_i), the dynamics in function space are

$$\frac{d}{dt} f(\theta, x) = -\sum_{i=1}^{N} K_\theta(x, x_i) \big(f_\theta(x_i, t) - y_i \big)$$

Conserved quantities in DLNs

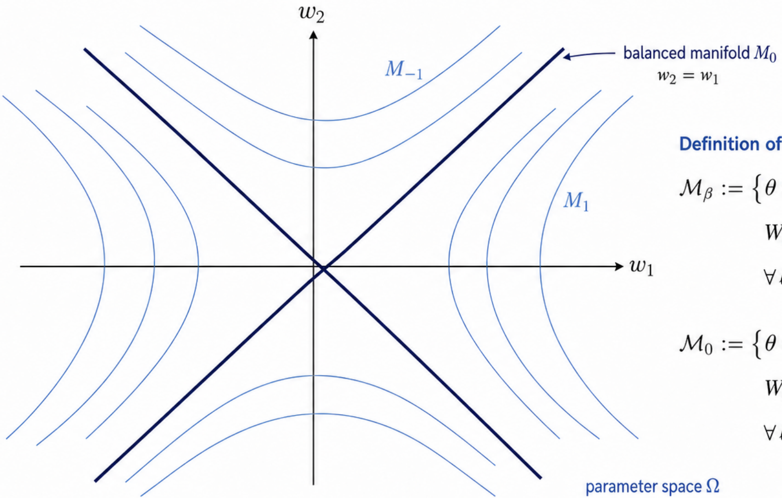
Conserved quantities under gradient flow:

$$G_l = W_{l+1}^\top W_{l+1} - W_l W_l^\top$$

When $\beta_l = 0$ we have **balanced weights**

For a 1D, 2-layer DLN, we get a hyperbolic equation: $w_2^2 - w_1^2 = \beta$

Foliation of parameter space by balance leaves



Definition of the balance leaves

$$\mathcal{M}_\beta := \Big\{ \theta \in \Omega : W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell, \quad \forall \ell = 1, \dots, L-1 \Big\}$$

$$\mathcal{M}_0 := \Big\{ \theta \in \Omega : W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = 0, \quad \forall \ell = 1, \dots, L-1 \Big\}$$

- Gradient flow remains on a fixed balance leaf.

The manifold of gradient flow

Gradient flow is constrained on the following hyperbolic manifold:

$$\mathcal{M}_\beta := \{\theta \in \Omega \mid W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell\}$$

Gradient flow on the balanced variety

On the balanced variety $\mathcal{M}_0$, the self-consistent gradient flow equation in function space is

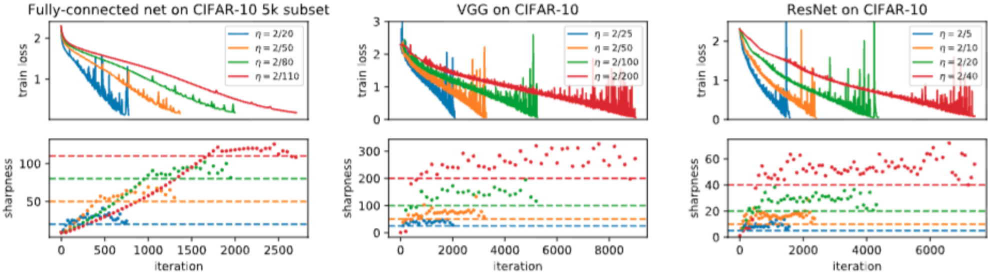
$$\dot{W} = \sum_{k=1}^{L} (WW^\top)^{\frac{L-k}{L}} R(W) (W^\top W)^{\frac{k-1}{L}}, \qquad R(W) := \Sigma_{YX} - W.$$

For depth 2:

$$\mathcal{M}_0 = \{(W_1, W_2) \mid W_2^\top W_2 = W_1 W_1^\top\}.$$

$$\dot{W} = (WW^\top)^{\frac{1}{2}} R(W) + R(W)(W^\top W)^{\frac{1}{2}}.$$

Descent lemma: For convex loss, loss decreases if $\eta < 2/L$
The loss is non-convex, but the Hessian oscillates slightly above $\frac{2}{\eta}$: it regularizes away sharp minima



Ref: Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, Cohen et al.

Takeaways

Key points

- NTK preconditions gradient flow in function space
- Symmetries can induce conserved quantites which constrain the gradient flow
- Gradient flow in function space for balanced quantities lead to self consistent equations
- The effect of learning rate can be understood via the edge of stability

References

- Menon, Govind. The geometry of the deep linear network. Symposium on Probability and Stochastic Processes. Cham: Springer Nature Switzerland, 2023.
- Sibylle Marcotte, Rémi Gribonval, and Gabriel Peyré. Abide by the law and follow the flow: Conservation laws for gradient flows. *Advances in neural information processing systems*, 36:63210–63221, 2023.
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- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. *Advances in neural information processing systems*, 31, 2018.
- Jeremy M Cohen, Alex Damian, Ameet Talwalkar, J Zico Kolter, and Jason D Lee. Understanding optimization in deep learning with central flows. arXiv preprint arXiv:2410.24206, 2024

Initialization, Width and Depth

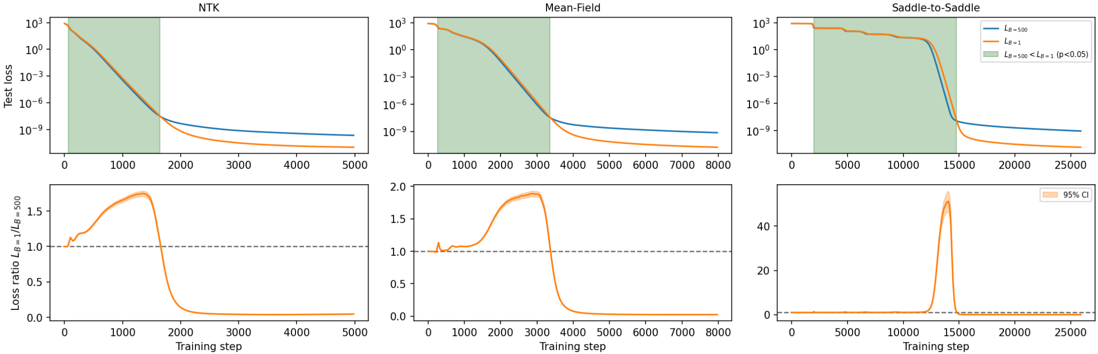
Training losses

Initialize weights: $\theta \sim \mathcal{N}(0, \sigma^{-\frac{\gamma}{2}})$

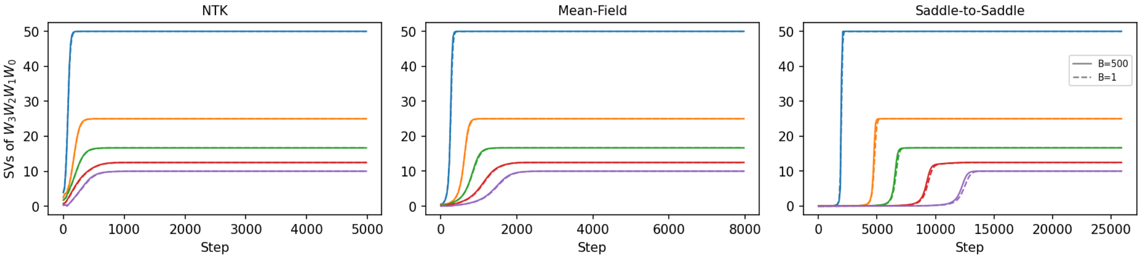
$\gamma < 1$: large initialization, linear regime (NTK)

$\gamma > 1$: small initialization, saddle-to-saddle

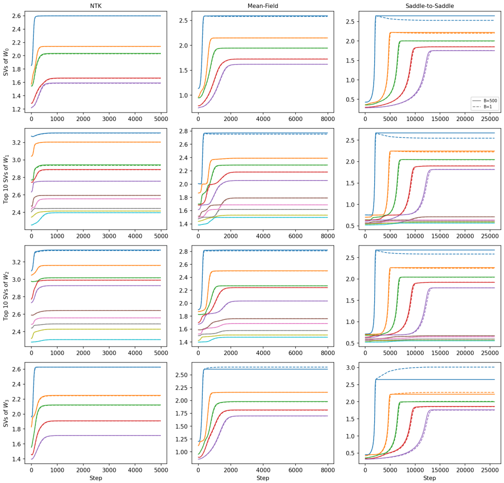
$\gamma = 1$: intermediate regime



Singular values of student matrix



Singular values of individual weight matrices



Lazy Regime (NTK or Linear regime)

For a DLN of depth L with initialization $\theta(0) \sim N(0, \sigma)$ and $\sigma^2 = d_L^{-\gamma}$, $\gamma < 1$:

$$\dot{W} \approx K_0[M - W]$$

Leading to the following solution:

$$W(t) = M - \exp(-K_0 t)[M - W(0)]$$

The time scale of learning is fixed by the NTK at initialization:

$$t \approx -k_0^{-1} \log\left(\frac{s - w_f}{s - w_0}\right)$$

Behaves like a shallow linear network

Rich regime

For balanced and aligned weights (or $\gamma > 1$), singular values of student satisfy:

$$\dot{w}_\alpha = 2\left(s_\alpha - w_\alpha\right) w_\alpha .$$

Solving this ODE for $w(t) := w_\alpha$ gives:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}} .$$

Timescale of learning

$$t = \frac{1}{2s} \ln\left(\frac{w_f\left(s - w_0\right)}{w_0\left(s - w_f\right)}\right), \qquad s \neq 0.$$

Takeaways

Key points

- Depth and specific initialization induce a sadde-to-saddle dynamics
- Gradient flow has a simplicity bias: low rank approximation
- Large initialization and width is NTK regime: no feature learning, NTK is frozen
- In saddle to saddle regime, time scale of learning set by the size of singular value of target function

References

- Saxe, Andrew M., James L. McClelland, and Surya Ganguli. A mathematical theory of semantic development in deep neural networks. *Proceedings of the National Academy of Sciences* 116.23 (2019): 11537-11546.
- Jacot, Arthur, et al. Saddle-to-saddle dynamics in deep linear networks: Small initialization training, symmetry, and sparsity. *arXiv preprint* arXiv:2106.15933 (2021).
- Dominé, Clémentine CJ, et al. From lazy to rich: Exact learning dynamics in deep linear networks. *arXiv preprint* arXiv:2409.14623 (2024)
- Zhang, Yedi, Andrew Saxe, and Peter E. Latham. Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures. *arXiv preprint* arXiv:2512.20607 (2025).
- Zhenfeng Tu, Santiago Tomas Aranguri Diaz, and Arthur Jacot. Mixed dynamics in linear networks: Unifying the lazy and active regimes. *Advances in Neural Information Processing Systems*, 37:106059–106104, 2024.

Stochasticity

Langevin dynamics

- SGD: Sample a batch from a dataset and compute the batch gradient
- Continuous model with B_t Brownian

$$d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{\eta\, \Sigma(\theta_t)}\, dB_t,$$

- Gradient noise covariance $\Sigma(\theta_t)$ depends on geometry: it is anisotropic and state-dependent

Fokker-Planck: Boltzmann equilibria

The Fokker-Planck equation (for density $p(\theta, t)$) is a local conservation law:

$$\begin{aligned} \partial_t p &:= -\nabla \cdot j \\ j &:= -\nabla L(\theta) p(\theta) - \frac{\eta}{2} \nabla^\top \left(\Sigma(\theta) p(\theta)\right) \end{aligned}$$

Important special case:

- Stationary distribution: $\partial_t p^*(x, t) = 0$
- Thermal equilibrium: $j = 0$
- White noise: $\Sigma = \sigma^2 I$

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta\sigma^2} L_N(\theta)\right) \text{ Boltzmann}$$

Noise anisotropy

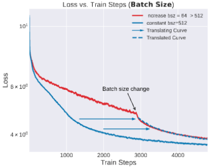
- In general, noise is anisotropic: it introduces a different implicit bias
- In function space, Langevin introduces a drift

$$\frac{\eta}{2}\,\mathrm{Tr}(\Sigma(\theta)H(\theta))$$

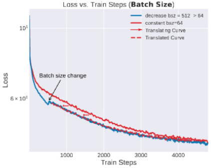
- Heuristic: bias toward flatter minima
- More:
  - see Govind Menon On the implicit regularization of Langevin dynamics with projected noise
  - My paper with collaborators deriving Langevin on DLNs during the saddle-to-saddle regime

Golden Path hypothesis

- Lots of debate about the implicit bias of SGD noise
- Transient (drift dominated) vs low-loss regime (diffusion dominated):



(a) Increasing Batch Size: C4



(b) Decreasing Batch Size: C4

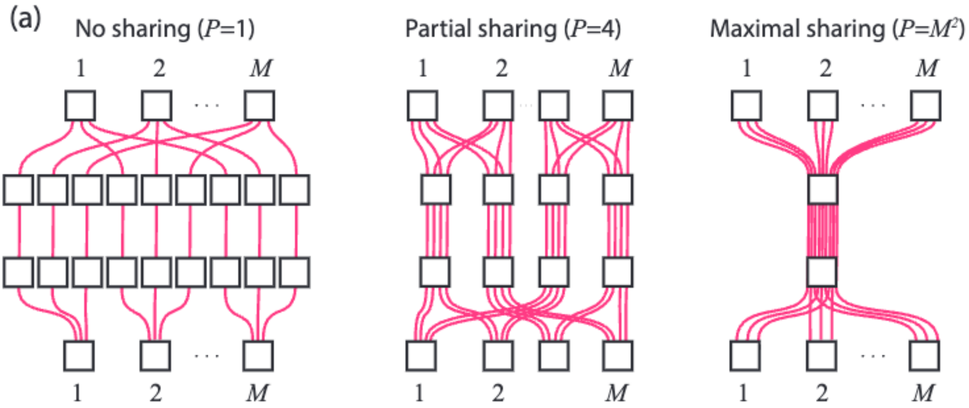
(c) Increasing Batch Size: CIFAR 5m

(d) Decreasing Batch Size: CIFAR 5m

Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Towards non linearity

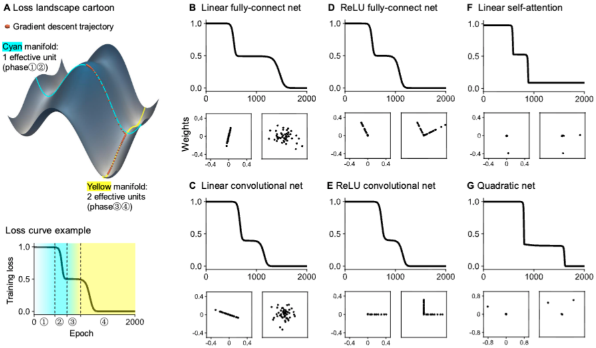
Neural race reduction



The Neural Race Reduction: Dynamics of Abstraction in Gated Networks, Andrew M. Saxe et al.

MLP and linear transformers

Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures, Saxe et al.



Takeaways

Key points

- SGD can be modelled via some SDE (with state dependent noise)
- If noise is white at thermal equilibrium the stationary distribution of the SDE is Boltzmann
- Implicit bias of Stochasticity not fully understood (flatness?) but more a 2nd order effect
- Non linearity can be modelled with gated DLN: neural race reduction
- MLP and linear transformers also have saddle to saddle dynamics

Promising directions:

- Polynomial neural networks (quadratic activations)
- Related: Tensor and tree tensor networks

Big questions

- Toy models of deep non linear learning
- Theory that capture natural data
- Making explicit the implicit biases of realistic Architectures
- Interplay between geometry and learning
- A statistical field theory of deep neural networks
- A more realistic theory of feature and hierarchical learning

Reference: *There Will Be a Scientific Theory of Deep Learning, Jamie Simon et al.*